EXP. NO: 25

SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB

Objectives

1. To generate a binary information sequence and encode it using a convolutional encoder in MATLAB.
2. To understand the relationship between the input bit sequence, encoder memory, code rate, and encoded output sequence.
3. To analyze the structure and length of the encoded output bit sequence for a selected convolutional code.
4. To visualize and interpret the input and encoded bit sequences and verify the encoding operation.
5. To study the effect of encoder parameters such as constraint length and generator polynomials on the encoded sequence.

CODE

```
clc;
clear;
close all;

% Generate input bits
N = 20;
data = randi([0 1], N, 1);

% Convolutional Encoder
% Rate = 1/2, Constraint Length = 3
% Generator polynomials: 7 and 5 (octal)

register = [0 0];
encoded = zeros(2*N, 1);

for i = 1:N
   b = data(i);

   % Shift register
   r1 = b;
   r2 = register(1);
   r3 = register(2);

   % Output bits using generator polynomials
   out1 = mod(r1 + r2 + r3, 2);
   out2 = mod(r1 + r3, 2);
```

```
    encoded(2*i-1) = out1;
    encoded(2*i) = out2;

    % Update register
    register = [b register(1)];
end

% Display results
disp('Input Bits:');
disp(data');

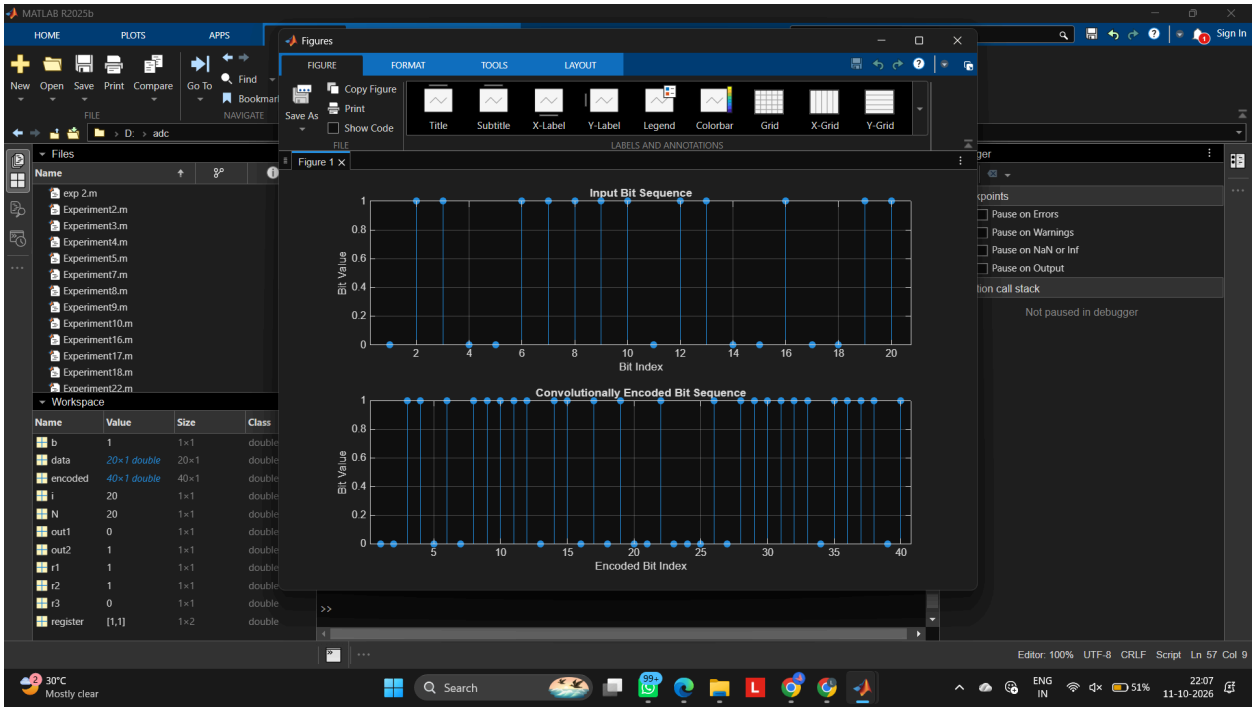
disp('Encoded Bits:');
disp(encoded');

% Plot results
figure;

subplot(2,1,1);
stem(1:N, data, 'filled');
title('Input Bit Sequence');
xlabel('Bit Index');
ylabel('Bit Value');
grid on;

subplot(2,1,2);
stem(1:2*N, encoded, 'filled');
title('Convolutionally Encoded Bit Sequence');
xlabel('Encoded Bit Index');
ylabel('Bit Value');
grid on;
```

OUTPUT


Input Bit Sequence
Bit Value
Bit Index
Convolutionally Encoded Bit Sequence
Bit Value
Encoded Bit Index